

```
1 //LAB CODE BAD MED
2 // JASON JOHNSON
3 // JOHNSON JASON
4 // 4/20/26
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6
7 #pragma once
8 #include <iostream>
9 using namespace std;
10
11 class FeetInches
12 {
13 private:
14     int feet;          // To hold a number of feet
15     int inches;        // To hold a number of inches
16     void simplify();   // Defined in FeetInches.cpp
17 public:
18     // Constructor
19     FeetInches(int f = 0, int i = 0);
20
21     // Mutator functions
22     void setFeet(int f)
23     {
24         feet = f;
25     }
26
27     void setInches(int i);
28
29
30     // Accessor functions
31     int getFeet() const
32     {
33         return feet;
34     }
35
36     int getInches() const
37     {
38         return inches;
39     }
40
41     // Overloaded operator functions
42     FeetInches operator + (const FeetInches&); // Overloaded +
43     FeetInches operator - (const FeetInches&); // Overloaded -
44     FeetInches operator ++ ();                // Prefix ++
45     FeetInches operator ++ (int);              // Postfix ++
46     bool operator > (const FeetInches&);       // Overloaded >
47     bool operator < (const FeetInches&);       // Overloaded <
48     bool operator == (const FeetInches&);      // Overloaded ==
49     // Friends
```

```
50     friend ostream& operator << (ostream&, const FeetInches&);
51     friend istream& operator >> (istream&, FeetInches&);
52 };
53
54 void FeetInches::simplify() {
55     int totalInches = (feet * 12) + inches;
56     feet = totalInches / 12;
57     inches = totalInches % 12;
58
59 }
60
61
62 FeetInches::FeetInches(int f, int i) {
63     feet = f;
64     inches = i;
65
66 }
67
68 void FeetInches::setInches(int i) {
69     inches = i;
70 }
71
72
73
74 FeetInches operator + (const FeetInches& first, FeetInches& second) {
75     return FeetInches(first.inches + second.getInches);
76
77 }
78
79 FeetInches operator - (const FeetInches& first, FeetInches& second) {
80     if (*this <) {
81
82     }
83 }
84
85 FeetInches operator ++ () {
86     return inches++;
87 }
88
89 FeetInches operator ++ (int i) {
90     return inches++;
91 }
92
93 bool operator > (const FeetInches& first, const FeetInches& second) {
94     return first > second;
95 }
96
97 bool operator < (const FeetInches& first, FeetInches& second ) {
98     return first < second;
```

```
99
100 }
101
102
103 bool operator == (const FeetInches& first, FeetInches& second) {
104     return first == second;
105 }
106
107 // Friends
108 ostream& operator << (ostream& stream, const FeetInches& f) {
109     stream <<
110 }
111
112 istream& operator >> (istream& stream, FeetInches& f) {
113
114
115 }
116
117
118
119
120
121
122
123
124
125
126
127
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135
136
137
138
139
140
141
142 // This program demonstrates the << and >> operators,
143 // overloaded to work with the FeetInches class.
144
145
146 int main()
147 {
```

```
148     FeetInches first, second; // Define two objects.
149
150     // Get a distance for the first object.
151     cout << "Enter a distance in feet and inches.\n";
152     cin >> first;
153
154     // Get a distance for the second object.
155     cout << "Enter another distance in feet and inches.\n";
156     cin >> second;
157
158     // Display the values in the objects.
159     cout << "\nThe values you entered are:\n\n";
160
161     cout << first << " and " << second << endl << endl;
162
163     if (first > second)
164     {
165         cout << "First is longer than the second.\n\n";
166     }
167     else
168     {
169         cout << "The second is longer than the first.\n\n";
170     }
171
172
173     if (first < second)
174     {
175         cout << "The Second is longer than the second.\n\n";
176     }
177     else
178     {
179         cout << "The First is longer than the first.\n\n";
180     }
181
182     if (first == second)
183     {
184         cout << "Both lengths are the same\n\n";
185     }
186     else
187     {
188         cout << "length are not the same.\n\n";
189     }
190
191
192
193     cout << "Adding first to second " << first + second << endl << endl;;
194
195     cout << "Subtracting first from second " << first - second << endl << ➤
        endl;
```

```
196
197     cout << "Incrementing first " << ++first << endl << endl;
198
199     cout << "Incrementing second " << ++second << endl << endl;
200
201     cout << "Incrementing first " << first++ << endl << endl;
202
203     cout << "Incrementing second " << second++ << endl << endl;
204
205
206
207     return 0;
208 }
209
210
211
```